

# MDP

$$S = \{s_1, \dots, s_n\}$$

$$A = \{a_1, \dots, a_n\}$$

$$A_L: S \rightarrow P(A)$$

$$S_T \subseteq S \text{ estados terminales}$$

$$T: S \times A \times S \rightarrow \mathbb{R} \quad T(s, a, s') = \Pr(S_{T+1} = s' \mid S_T = s, A_T = a)$$

$$r: S \times A \times S \rightarrow \mathbb{R} \quad r(s, a, s') \quad \sum_{s' \in S} T(s, a, s') = 1$$

$$\gamma: \gamma \in [0, 1] \text{ Factor de descuento}$$

$$\pi: S \times A \rightarrow \mathbb{R} \quad \pi(s, a) = \Pr(A_T = a \mid S_T = s) \quad \sum_{a \in A_T(s)} \pi(s, a) = 1$$

$$\pi^* = \arg \max_{\pi} E^{\pi}[R_T] \quad \forall s \in S$$
$$E^{\pi}[r_T + \gamma r_{T+1} + \gamma^2 r_{T+2} + \dots] \quad \forall s \in S$$
$$E^{\pi}[r_T + \gamma R_{T+1}] \quad \forall s \in S$$

$$V^{\pi}(s) = E^{\pi}[r_T + \gamma R_{T+1} \mid S_T = s] = \sum_{a \in A_L(s)} \pi(s, a) \sum_{s' \in S} T(s, a, s') [r(s, a, s') + \gamma V^{\pi}(s')]$$

$$V^*(s) = \max_{a \in A_L(s)} \sum_{s' \in S} T(s, a, s') [r(s, a, s') + \gamma V^*(s')]$$

$$\pi^*(s, a) = \begin{cases} 1 & \text{si } a = \arg \max_{a'} \sum_{s' \in S} T(s, a', s') [r(s, a', s') + \gamma V^*(s')] \\ 0 & \text{en otro caso} \end{cases}$$

## MDPsim

$$S = \{s_1, \dots, s_n\}$$

$$A = \{a_1, \dots, a_n\}$$

$$A_L: S \rightarrow P(A)$$

$$S_T \subseteq S$$

$$s' = \text{Successor}(s, a)$$

$$r: S \times A \times S \rightarrow \mathbb{R} \quad r(s, a, s')$$

$$\gamma = \gamma \in [0, 1]$$

¿Cómo evaluar una política  $\pi$  con MDPsim?

$$V^\pi(s) = \sum_{a \in A_s} \pi(s, a) \sum_{s' \in S} T(s, a, s') [r(s, a, s') + \gamma V^\pi(s')]$$

asumo que conozco  $\hat{V}^\pi(s)$  y estoy en el estado  $s$   
escojo  $a$  a partir de  $\pi$  aplico  $a$  a MDPsim y obtengo  $s'$  y  $r$   
 $s' = \text{sucesor}(s, a)$   
 $r = r(s, a, s')$

entonces  $\hat{V}_{t+1}^\pi(s) = r + \gamma \hat{V}^\pi(s')$

$$\hat{V}^\pi(s) \leftarrow \hat{V}^\pi(s) + \alpha [\hat{V}_{t+1}^\pi(s) - \hat{V}^\pi(s)]$$

Diferencia Temporal  $\Delta$

$\pi$  determinista:

$$a = \pi(s)$$

$\pi$  estocástico:

$$\text{umbral} = \text{random} \cdot \text{random}()$$

$$\text{nivel} = 0$$

For  $a$  in  $A_L(s)$ :

$$\text{nivel} += \pi(s, a)$$

if  $\text{umbral} \leq \text{nivel}$ :

return  $a$

raise ValueError(—)